Renal, Electrolytes & RRT – Chapter Summary

AKI • Electrolytes • Renal Replacement Therapy

ICU / EDIC Board Review • MCQ-Oriented

1. AKI Definition (KDIGO)

Stage	Creatinine	Urine output
1	↑ ≥ 0.3 mg/dL in 48 h or 1.5–1.9 × baseline	< 0.5 mL/kg/h × 6–12 h
2	2.0–2.9 × baseline	< 0.5 mL/kg/h ≥ 12 h
3	≥ 3 × or ≥ 4 mg/dL or RRT	< 0.3 mL/kg/h ≥ 24 h or anuria ≥ 12 h

- Creatinine lags; ↓ muscle mass & fluid overload underestimate AKI
- **Biomarkers:** NGAL, TIMP-2 × IGFBP7, cystatin C
- **Furosemide stress test:** UO < 200 mL in 2 h after 1–1.5 mg/kg → progression risk

2. AKI Causes & Workup

	Prerenal	ATN
Urine Na	< 20	> 40
FENa	< 1%	> 2%
FEUrea (on diuretics)	< 35%	> 50%
Urine osm	> 500	< 350
Sediment	Bland	Muddy brown casts

- **Postrenal:** ultrasound for hydronephrosis
- **RBC casts** → glomerulonephritis; **WBC casts** → interstitial nephritis / pyelonephritis
- **Eosinophiluria** → AIN (β-lactams, PPI, NSAIDs)
- **Sepsis** = most common cause of ICU AKI

3. AKI Prevention & Management

- **KDIGO bundle:** stop nephrotoxins, optimize volume & MAP, avoid hyperglycemia, monitor creatinine & UO
- **Balanced crystalloids** over saline (SMART)
- **Contrast:** IV hydration if high risk; no benefit of NAC / bicarbonate (PRESERVE)
- **Diuretics** don't prevent / treat AKI – only for volume overload
- **Fluid overload > 10%** = ↑ mortality
- **Dose adjust** drugs; check aminoglycoside / vancomycin levels

4. Renal Replacement Therapy – Timing

Emergency indications (AEIOU): Acidosis (pH < 7.15), Electrolytes (K > 6–6.5 refractory), Intoxication (dialyzable), Overload (refractory pulmonary edema), Uremia (pericarditis, encephalopathy, bleeding)

- **Accelerated vs standard start:** no benefit (STARRT-AKI, AKIKI, IDEAL-ICU)
- **AKIKI 2:** "more delayed" start (BUN > 140 mg/dL) → harm
- **ELAIN** (early, surgical): benefit – single center
- Watchful waiting is reasonable without urgent indication

5. RRT Modalities & Dose

Modality	Mechanism	Best for
CVVH	Convection	Middle molecules
CVVHD	Diffusion	Small solutes
CVVHDF	Both	Common ICU choice
IHD	Diffusion, high flow	Stable patients, toxins, ↑ K rapidly
SLED	Hybrid	Intermediate stability
PD	Peritoneal	Resource-limited

- **CRRT = IHD for survival**; CRRT better for hemodynamic instability & raised ICP / ALF
- **Dose:** effluent **20–25 mL/kg/h** delivered (prescribe 25–30) (RENAL, ATN trials)
- **Filtration fraction** < 20–25% (pre-dilution ↓ clotting)
- **Catheter:** right IJ > femoral > left IJ; avoid subclavian

6. Anticoagulation for CRRT

- **Regional citrate** = first-line (KDIGO) – ↑ filter life, ↓ bleeding
- **Citrate accumulation:** total/ionized Ca ratio > 2.5, ↑ AG metabolic acidosis, ↓ iCa (liver failure, shock)
- **Citrate excess delivery:** metabolic alkalosis
- **Heparin** if citrate contraindicated; none if high bleeding risk
- Post-filter iCa 0.25–0.35; systemic iCa 1.0–1.2

7. Hyponatremia

- **Severe symptoms** (seizure, coma): **3% NaCl 150 mL over 20 min** × 2–3 → target ↑ 4–6 mmol/L
- **Correction limit:** ≤ 8–10 mmol/L/24 h (≤ 8 in high risk: alcohol, malnutrition, hypokalemia, liver disease)
- **Overcorrection:** D5W ± desmopressin (reverse)
- **Hypovolemic** (urine Na < 30): saline; **euvolemic** (SIADH: urine osm > 100, urine Na > 30): fluid restriction, urea, tolvaptan; **hypervolemic:** HF, cirrhosis
- **Cerebral salt wasting** (SAH): hypovolemia + natriuresis → saline + fludrocortisone
- **Pseudohyponatremia:** lipids, proteins; **hypertonic:** glucose (correct Na +1.6–2.4 per 100 mg/dL)

8. Hypernatremia & Diabetes Insipidus

- **Free water deficit** = TBW × (Na/140 – 1)
- Correct ≤ 10–12 mmol/L/24 h if chronic; acute (< 48 h) can be faster
- **Central DI:** urine osm < 300, polyuria → desmopressin (↑ urine osm > 50%)
- **Nephrogenic DI:** lithium, hypercalcemia, hypokalemia → thiazide, amiloride (lithium)
- **Brain death / pituitary surgery:** central DI common

9. Potassium

Hyperkalemia

- **ECG:** peaked T → PR ↑, P flat → QRS wide → sine wave → VF
- **Stabilize:** calcium gluconate 10% 30 mL (or chloride 10 mL) – repeat if ECG persists
- **Shift:** insulin 10 U + glucose 25 g; salbutamol 10–20 mg neb; bicarbonate if acidotic
- **Remove:** loop diuretics, SZC / patiromer, dialysis

Hypokalemia

- U waves, flat T, arrhythmias; correct **Mg** first if low
- IV max 10–20 mmol/h peripheral (≤ 40 via central with monitoring)

10. Calcium, Magnesium & Phosphate

- **Corrected Ca** = Ca + 0.8 × (4 – albumin g/dL)
- **Hypercalcemia:** saline 200–300 mL/h, **calcitonin** (fast, tachyphylaxis), **zoledronate** (2–4 days), denosumab (renal failure), dialysis; steroids in granulomatous / lymphoma
- **Hypocalcemia:** symptomatic → calcium gluconate IV; check Mg
- **Hypomagnesemia:** refractory hypoK/hypoCa; torsades → 2 g IV
- **Hypophosphatemia:** refeeding syndrome → weakness, respiratory failure, hemolysis; replace if < 0.6 mmol/L
- **Refeeding:** thiamine, start 10–20 kcal/kg/day, monitor PO_4, K, Mg

11. Acid–Base in Renal Disease

- **Normal AG (hyperchloremic):** diarrhea, RTA, saline, ureteral diversion
- **Urine AG** = UNa + UK – UCl: negative = GI loss (↑ NH_4^+); positive = RTA

RTA	Defect	K	Urine pH
Type 1 (distal)	H^+ secretion	↓	> 5.5
Type 2 (proximal)	HCO_3 reabsorption	↓	< 5.5 (when low HCO_3)
Type 4	Hypoaldosteronism	↑	< 5.5

- **Bicarbonate therapy:** pH ≤ 7.20 + AKIN 2–3 (BICAR-ICU) → ↓ RRT & mortality

12. Rhabdomyolysis

- CK > 5× ULN; risk of AKI if CK > 5000–15000
- **Causes:** trauma / crush, exertion, seizures, drugs (statins, propofol), hyperthermia, compartment syndrome
- **Rx:** aggressive crystalloid (UO 200–300 mL/h); bicarbonate / mannitol: no proven benefit
- Hyperkalemia, hypocalcemia (early), hypercalcemia (recovery), hyperphosphatemia

13. Special Renal Syndromes

- **Hepatorenal syndrome (HRS-AKI):** cirrhosis + AKI not responding to albumin 1 g/kg × 2 days → **terlipressin + albumin** (CONFIRM); NE alternative
- **Abdominal compartment syndrome:** IAP > 20 + new organ failure → AKI (decompress)
- **Cardiorenal:** venous congestion → decongest
- **Contrast nephropathy:** overstated risk; hydrate high-risk patients
- **TLS:** see oncology – rasburicase, hydration

14. Toxins & Extracorporeal Removal

- **Dialyzable ("I STUMBLE"):** Isopropanol, Salicylates, Theophylline, Uremia, Methanol, Barbiturates, Lithium, Ethylene glycol
- **Not dialyzable:** high Vd / protein-bound (digoxin, TCAs, β-blockers mostly)
- **IHD preferred** for toxins (highest clearance)
- **Toxic alcohols:** fomepizole; dialysis if acidosis / end-organ damage
- **Lithium:** dialysis if level > 4 (> 2.5 with neuro symptoms / renal impairment)
- **Metformin-associated lactic acidosis:** dialysis if lactate > 20, pH ≤ 7.0

15. High-Yield MCQ Pearls

- [x] KDIGO stage 1: Cr ↑ 0.3 in 48 h
- [x] FEUrea < 35% = prerenal on diuretics
- [x] Early RRT start: no benefit (STARRT-AKI)
- [x] Effluent dose 20–25 mL/kg/h delivered
- [x] Citrate accumulation: total/ionized Ca > 2.5
- [x] Na correction ≤ 8–10/24 h
- [x] Hyperkalemia: calcium first (membrane)
- [x] Correct Mg before K
- [x] HRS: terlipressin + albumin
- [x] Hypercalcemia: saline + calcitonin + bisphosphonate
- [x] Lithium & toxic alcohols → IHD
- [x] Bicarbonate if pH ≤ 7.2 + AKI

1. AKI Definition (KDIGO)

Stage	Creatinine	Urine output
1	↑ ≥ 0.3 mg/dL in 48 h or 1.5–1.9 × baseline	< 0.5 mL/kg/h × 6–12 h
2	2.0–2.9 × baseline	< 0.5 mL/kg/h ≥ 12 h
3	≥ 3 × or ≥ 4 mg/dL or RRT	< 0.3 mL/kg/h ≥ 24 h or anuria ≥ 12 h

- Creatinine lags; ↓ muscle mass & fluid overload underestimate AKI
- **Biomarkers:** NGAL, TIMP-2 × IGFBP7, cystatin C
- **Furosemide stress test:** UO < 200 mL in 2 h after 1–1.5 mg/kg → progression risk

2. AKI Causes & Workup

	Prerenal	ATN
Urine Na	< 20	> 40
FENa	< 1%	> 2%
FEUrea (on diuretics)	< 35%	> 50%
Urine osm	> 500	< 350
Sediment	Bland	Muddy brown casts

- **Postrenal:** ultrasound for hydronephrosis
- **RBC casts** → glomerulonephritis; **WBC casts** → interstitial nephritis / pyelonephritis
- **Eosinophiluria** → AIN (β-lactams, PPI, NSAIDs)
- **Sepsis** = most common cause of ICU AKI

3. AKI Prevention & Management

- **KDIGO bundle:** stop nephrotoxins, optimize volume & MAP, avoid hyperglycemia, monitor creatinine & UO
- **Balanced crystalloids** over saline (SMART)
- **Contrast:** IV hydration if high risk; no benefit of NAC / bicarbonate (PRESERVE)
- **Diuretics** don't prevent / treat AKI – only for volume overload
- **Fluid overload > 10%** = ↑ mortality
- **Dose adjust** drugs; check aminoglycoside / vancomycin levels

4. Renal Replacement Therapy – Timing

Emergency indications (AEIOU): Acidosis (pH < 7.15), Electrolytes (K > 6–6.5 refractory), Intoxication (dialyzable), Overload (refractory pulmonary edema), Uremia (pericarditis, encephalopathy, bleeding)

- **Accelerated vs standard start:** no benefit (STARRT-AKI, AKIKI, IDEAL-ICU)
- **AKIKI 2:** "more delayed" start (BUN > 140 mg/dL) → harm
- **ELAIN** (early, surgical): benefit – single center
- Watchful waiting is reasonable without urgent indication

5. RRT Modalities & Dose

Modality	Mechanism	Best for
CVVH	Convection	Middle molecules
CVVHD	Diffusion	Small solutes
CVVHDF	Both	Common ICU choice
IHD	Diffusion, high flow	Stable patients, toxins, ↑ K rapidly
SLED	Hybrid	Intermediate stability
PD	Peritoneal	Resource-limited

- **CRRT = IHD for survival**; CRRT better for hemodynamic instability & raised ICP / ALF
- **Dose:** effluent **20–25 mL/kg/h** delivered (prescribe 25–30) (RENAL, ATN trials)
- **Filtration fraction** < 20–25% (pre-dilution ↓ clotting)
- **Catheter:** right IJ > femoral > left IJ; avoid subclavian

6. Anticoagulation for CRRT

- **Regional citrate** = first-line (KDIGO) – ↑ filter life, ↓ bleeding
- **Citrate accumulation:** total/ionized Ca ratio > 2.5, ↑ AG metabolic acidosis, ↓ iCa (liver failure, shock)
- **Citrate excess delivery:** metabolic alkalosis
- **Heparin** if citrate contraindicated; none if high bleeding risk
- Post-filter iCa 0.25–0.35; systemic iCa 1.0–1.2

7. Hyponatremia

- **Severe symptoms** (seizure, coma): **3% NaCl 150 mL over 20 min** × 2–3 → target ↑ 4–6 mmol/L
- **Correction limit:** ≤ 8–10 mmol/L/24 h (≤ 8 in high risk: alcohol, malnutrition, hypokalemia, liver disease)
- **Overcorrection:** D5W ± desmopressin (reverse)
- **Hypovolemic** (urine Na < 30): saline; **euvolemic** (SIADH: urine osm > 100, urine Na > 30): fluid restriction, urea, tolvaptan; **hypervolemic:** HF, cirrhosis
- **Cerebral salt wasting** (SAH): hypovolemia + natriuresis → saline + fludrocortisone
- **Pseudohyponatremia:** lipids, proteins; **hypertonic:** glucose (correct Na +1.6–2.4 per 100 mg/dL)

8. Hypernatremia & Diabetes Insipidus

- **Free water deficit** = TBW × (Na/140 − 1)
- Correct ≤ 10–12 mmol/L/24 h if chronic; acute (< 48 h) can be faster
- **Central DI:** urine osm < 300, polyuria → desmopressin (↑ urine osm > 50%)
- **Nephrogenic DI:** lithium, hypercalcemia, hypokalemia → thiazide, amiloride (lithium)
- **Brain death / pituitary surgery:** central DI common

9. Potassium

Hyperkalemia

- **ECG:** peaked T → PR ↑, P flat → QRS wide → sine wave → VF
- **Stabilize:** calcium gluconate 10% 30 mL (or chloride 10 mL) – repeat if ECG persists
- **Shift:** insulin 10 U + glucose 25 g; salbutamol 10–20 mg neb; bicarbonate if acidotic
- **Remove:** loop diuretics, SZC / patiromer, dialysis

Hypokalemia

- U waves, flat T, arrhythmias; correct **Mg** first if low
- IV max 10–20 mmol/h peripheral (≤ 40 via central with monitoring)

10. Calcium, Magnesium & Phosphate

- **Corrected Ca** = Ca + 0.8 × (4 − albumin g/dL)
- **Hypercalcemia:** saline 200–300 mL/h, **calcitonin** (fast, tachyphylaxis), **zoledronate** (2–4 days), denosumab (renal failure), dialysis; steroids in granulomatous / lymphoma
- **Hypocalcemia:** symptomatic → calcium gluconate IV; check Mg
- **Hypomagnesemia:** refractory hypoK/hypoCa; torsades → 2 g IV
- **Hypophosphatemia:** refeeding syndrome → weakness, respiratory failure, hemolysis; replace if < 0.6 mmol/L
- **Refeeding:** thiamine, start 10–20 kcal/kg/day, monitor PO_4, K, Mg

11. Acid–Base in Renal Disease

- **Normal AG (hyperchloremic):** diarrhea, RTA, saline, ureteral diversion
- **Urine AG** = UNa + UK − UCl: negative = GI loss (↑ NH_4^+); positive = RTA

RTA	Defect	K	Urine pH
Type 1 (distal)	H^+ secretion	↓	> 5.5
Type 2 (proximal)	HCO_3 reabsorption	↓	< 5.5 (when low HCO_3)
Type 4	Hypoaldosteronism	↑	< 5.5

- **Bicarbonate therapy:** pH ≤ 7.20 + AKIN 2–3 (BICAR-ICU) → ↓ RRT & mortality

12. Rhabdomyolysis

- CK > 5× ULN; risk of AKI if CK > 5000–15000
- **Causes:** trauma / crush, exertion, seizures, drugs (statins, propofol), hyperthermia, compartment syndrome
- **Rx:** aggressive crystalloid (UO 200–300 mL/h); bicarbonate / mannitol: no proven benefit
- Hyperkalemia, hypocalcemia (early), hypercalcemia (recovery), hyperphosphatemia

13. Special Renal Syndromes

- **Hepatorenal syndrome (HRS-AKI):** cirrhosis + AKI not responding to albumin 1 g/kg × 2 days → **terlipressin + albumin** (CONFIRM); NE alternative
- **Abdominal compartment syndrome:** IAP > 20 + new organ failure → AKI (decompress)
- **Cardiorenal:** venous congestion → decongest
- **Contrast nephropathy:** overstated risk; hydrate high-risk patients
- **TLS:** see oncology – rasburicase, hydration

14. Toxins & Extracorporeal Removal

- **Dialyzable ("I STUMBLE"):** Isopropanol, Salicylates, Theophylline, Uremia, Methanol, Barbiturates, Lithium, Ethylene glycol
- **Not dialyzable:** high Vd / protein-bound (digoxin, TCAs, β-blockers mostly)
- **IHD preferred** for toxins (highest clearance)
- **Toxic alcohols:** fomepizole; dialysis if acidosis / end-organ damage
- **Lithium:** dialysis if level > 4 (> 2.5 with neuro symptoms / renal impairment)
- **Metformin-associated lactic acidosis:** dialysis if lactate > 20, pH ≤ 7.0

15. High-Yield MCQ Pearls

- ✔ KDIGO stage 1: Cr ↑ 0.3 in 48 h
- ✔ FEUrea < 35% = prerenal on diuretics
- ✔ Early RRT start: no benefit (STARRT-AKI)
- ✔ Effluent dose 20–25 mL/kg/h delivered
- ✔ Citrate accumulation: total/ionized Ca > 2.5
- ✔ Na correction ≤ 8–10/24 h
- ✔ Hyperkalemia: calcium first (membrane)
- ✔ Correct Mg before K
- ✔ HRS: terlipressin + albumin
- ✔ Hypercalcemia: saline + calcitonin + bisphosphonate
- ✔ Lithium & toxic alcohols → IHD
- ✔ Bicarbonate if pH ≤ 7.2 + AKI